MASTERFUNCTIONALSKILLS

★ OFQUAL REGULATED STANDARDS

FUNCTIONAL SKILLS MATHEMATICS — Level 2 Specialist

Maths Level 2 Revision Cards

Visual Rules • Exam Challenges • Step-by-Step Solutions

- ★ Shapes, Circles, Trapeziums & Cylinders
- ★ Reverse Percentages & Compound Interest
- ★ Reverse Ratios & Grouped Frequency Mean
- ★ Step-by-Step Pearson & C&G Model Answers

A6 Postcard Cardstock (148 × 105 mm)

Turn over for Deck Contents & Checklist ➔

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1. Order of Operations (BIDMAS)

NUMBER SYSTEM

B - I - D/M - A/S

Operations must be completed in order: 1. Brackets (), 2. Indices (², ³, √), 3. Division & Multiplication (left-to-right), 4. Addition & Subtraction (left-to-right).

B | I | D | M | A | S

Example: 5 + 3 × (4 - 2)²

1. (4-2)=2 → 2. 2²=4 → 3. 3×4=12 → 4. 5+12 = 17

★ Golden Rule: Always solve brackets and powers first before multiplying or adding!

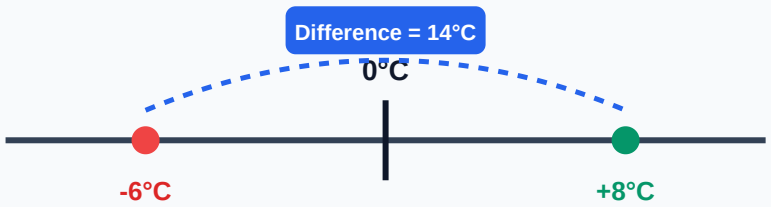
⚠️ Watch Out: Don't just calculate left-to-right! 5 + 3 × 2 is 11, NOT 16.

2. Negative Numbers in Context

NUMBER SYSTEM

Rules of Signs & Difference

Difference between positive & negative = Add their absolute values (|a| + |b|). Multiplying/Dividing: Same signs = POSITIVE (+). Different signs = NEGATIVE (-).



(-6) to (+8) = 6 steps to zero + 8 steps = 14

★ Golden Rule: Subtracting a negative is the same as ADDING! 5 - (-3) = 5 + 3 = 8.

⚠️ Watch Out: -5°C is COLDER than -2°C. The larger the numeral after the minus sign, the lower the value.

3. Fractions, Decimals & % Equivalence

NUMBER SYSTEM

Conversion Benchmarks

To compare values, convert them all to percentages or decimals: Fraction to %: (Numerator ÷ Denominator) × 100. Decimal to %: Multiply by 100.

Fraction	Decimal	Percentage
1/8	0.125	12.5%
3/8	0.375	37.5%
5/8	0.625	62.5%

★ Golden Rule: Eighths are common exam traps: 1/8 = 12.5% (0.125), not 18%!

⚠️ Watch Out: 0.4 is 40%, NOT 4%! 0.04 is 4%. Count your decimal places carefully.

Challenge: Order of Operations (BIDMAS)

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N1

Evaluate: 8 + 4 × (7 - 4)² ÷ 6

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1. Brackets: (7 - 4) = 3. 2. Index: 3² = 9. 3. Multiply: 4 × 9 = 36. 4. Divide: 36 ÷ 6 = 6. 5. Add: 8 + 6 = 14.

📋 DECK CONTENTS & REVISION CHECKLIST

50 Revision Cards

Tick [✓] Read when you understand the rule, and [★] Solved when you answer the challenge! Pearson & C&G Aligned

1. Order of Operations (BIDMAS) (DfE L2.N1)
2. Negative Numbers in Context (DfE L2.N2)
3. Fractions, Decimals & % Equivalence (DfE L2.N3)
4. Fraction of an Amount (Non-Calc) (DfE L2.N4)
5. Adding & Subtracting Mixed Fractions (DfE L2.N4)
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7. Percentage Increase & Decrease (Multiplier) (DfE L2.N6)
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19. Perimeter of Composite Shapes (DfE L2.MSS18)
20. Circumference of a Circle (DfE L2.MSS19)
21. Area of Triangle & Parallelogram (DfE L2.MSS20)
22. Area of a Trapezium (DfE L2.MSS20)
23. Area of a Circle & Semicircle (DfE L2.MSS21)
24. Area of Composite Shapes (DfE L2.MSS22)
25. Volume of Cuboids & Prisms (DfE L2.MSS23)
26. Volume of a Cylinder (DfE L2.MSS24)
27. Surface Area of a Cuboid (DfE L2.MSS25)
28. Surface Area of a Cylinder (DfE L2.MSS25)
29. Metric Unit Conversions (DfE L2.MSS26)
30. Converting Square & Cubic Units (DfE L2.MSS27)
31. Metric to Imperial Approximations (DfE L2.MSS28)
32. Speed, Distance & Time (DfE L2.MSS29)
33. Density, Mass & Volume (DfE L2.MSS30)
34. Scale Drawings & Maps (DfE L2.MSS31)
35. Bearings (3-Figure Bearings) (DfE L2.MSS32)
36. Plans & Elevations (DfE L2.MSS33)
37. Averages: Mean, Median, Mode & Range (DfE L2.HD34)
38. Mean from a Frequency Table (DfE L2.HD35)
39. Estimated Mean (Grouped Data) (DfE L2.HD36)
40. Modal Class & Median Group (DfE L2.HD37)
41. Comparing Two Data Sets (DfE L2.HD38)
42. Probability of Single Events (DfE L2.HD39)
43. Combined Independent Events (DfE L2.HD40)
44. Probability Tree Diagrams (DfE L2.HD41)
45. Two-Way Tables & Probability (DfE L2.HD42)
46. Scatter Diagrams & Correlation (DfE L2.HD43)
47. Interpolation vs Extrapolation (DfE L2.HD44)
48. Pie Charts: Angles & Frequencies (DfE L2.HD45)
49. Dual & Grouped Bar Charts (DfE L2.HD46)
50. Misleading Graphs & Traps (DfE L2.HD47)

💡 **Tutor Golden Tip (Abdul):** Do not flip the card immediately! Attempt the Quick Challenge on the back using a dry-wipe pen or scratchpad first. Verify with the step-by-step model steps only after completing your working.

Challenge: Fractions, Decimals & % Equivalence

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N3

Put in order from smallest to largest: 3/8, 35%, 0.36, 1/3.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Convert all to decimals: 1/3 = 0.333..., 35% = 0.35, 0.36 = 0.36, 3/8 = 0.375. Order: 1/3, 35%, 0.36, 3/8.

Challenge: Negative Numbers in Context

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N2

A vaccine freezer is at -18°C. A power failure causes it to rise to -4°C. By how many degrees did the temperature rise?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Rise = -4 - (-18) = -4 + 18 = 14°C.

4. Fraction of an Amount (Non-Calc)

NUMBER SYSTEM

Divide by Bottom, Times by Top

To find a fraction (a/b) of any amount: 1. Divide total by denominator (b) to find 1 part. 2. Multiply result by numerator (a) to get required parts.

Find 3/5 of £180

Step 1: Divide by 5

£180 ÷ 5 = £36 (1 part)

Step 2: Multiply by 3

£36 × 3 = £108

⋆ Golden Rule: Divide by the denominator, multiply by the numerator!

⚠️ Watch Out: If a question asks for the REMAINING amount, remember to subtract your answer from the original total.

5. Adding & Subtracting Mixed Fractions

NUMBER SYSTEM

Improper & Common Denominator

1. Convert mixed numbers to improper fractions (Whole × Denom + Num). 2. Find lowest common denominator (LCM). 3. Add or subtract numerators only. 4. Simplify.

2 1/4 + 1 2/3

1. Improper: 9/4 + 5/3
2. Common Denom (12): 27/12 + 20/12
3. Add numerators: 47/12 = 3 11/12

⋆ Golden Rule: NEVER add denominators together! 1/4 + 2/3 is NEVER 3/7.

⚠️ Watch Out: When subtracting, always convert to top-heavy improper fractions first to prevent negative fractional parts.

6. Multiplying & Dividing Fractions

NUMBER SYSTEM

Multiply Straight / Keep-Change-Flip

Multiplication: Multiply tops, multiply bottoms (cross-cancel if possible). Division: KEEP the first fraction, CHANGE ÷ to ×, FLIP the second fraction (reciprocal).

MULTIPLY

2/3 × 5/7

Top: 2 × 5 = 10

Bottom: 3 × 7 = 21 → 10/21

DIVIDE (K-C-F)

3/4 ÷ 2/5

Keep 3/4 • Change × • Flip 5/2

3/4 × 5/2 = 15/8 = 1 7/8

⋆ Golden Rule: Dividing by a fraction is the same as multiplying by its reciprocal!

⚠️ Watch Out: Never flip the FIRST fraction — only flip the SECOND fraction after changing the sign.

7. Percentage Increase & Decrease (Multiplier)

NUMBER SYSTEM

Single Multiplier Method

Increase: Add % to 100%, divide by 100. (e.g. +17.5% = × 1.175). Decrease: Subtract % from 100%, divide by 100. (e.g. -15% = × 0.85).

INCREASE (+20%)

100% + 20% = 120%

Multiplier = × 1.20

DECREASE (-15%)

100% - 15% = 85%

Multiplier = × 0.85

⋆ Golden Rule: Using a single multiplier is faster and eliminates rounding errors on calculator papers!

⚠️ Watch Out: Be careful with single digit percentages: +5% is × 1.05, NOT × 1.5 (which is +50%)!

Challenge: Adding & Subtracting Mixed Fractions

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N4

Work out 3 1/3 - 1 3/4. Give your answer as a mixed number in simplest form.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1. Convert: 10/3 - 7/4. 2. Common denom 12: 40/12 - 21/12 = 19/12. 3. Mixed: 1 7/12.

Challenge: Fraction of an Amount (Non-Calc)

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N4

A clinic orders 240 boxes of gloves. 5/8 of them are size Medium. How many boxes are NOT size Medium?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Non-Medium fraction = 1 - 5/8 = 3/8. 240 ÷ 8 = 30. 30 × 3 = 90 boxes.

Challenge: Percentage Increase & Decrease (Multiplier)

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N6

A laptop originally costs £640. In a sale, it is reduced by 18%. Use a single multiplier to calculate the sale price.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Multiplier = 1 - 0.18 = 0.82. Sale price = £640 × 0.82 = £524.80.

Challenge: Multiplying & Dividing Fractions

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N5

Work out 2/5 ÷ 4/15. Give your answer in its simplest form.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Keep 2/5, change to ×, flip to 15/4: (2 × 15) ÷ (5 × 4) = 30 ÷ 20 = 3/2 = 1 1/2.

8. Percentage Change Formula

NUMBER SYSTEM

(Change ÷ Original) × 100

Percentage Change = (Actual Difference ÷ Original Starting Value) × 100. Works for percentage profit, loss, discount, and inflation.

% Change = (Change ÷ Original) × 100

Always divide by the STARTING value, never the new value!

Car bought for £8,000, sold for £6,800

Change = £1,200 → (1,200 ÷ 8,000) × 100 = 15% Loss

⋆ Golden Rule: Denominator is ALWAYS the original starting value before the change occurred!

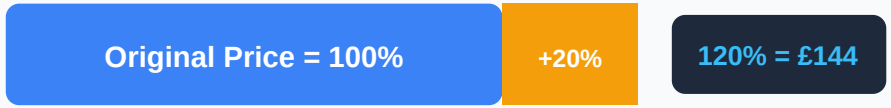
⚠ Watch Out: Don't divide by the new price! Dividing 1,200 by 6,800 gives 17.6%, which is incorrect.

9. Reverse Percentages (Original Value)

NUMBER SYSTEM

Divide by Multiplier

To find the original price before VAT or sale: 1. Identify the new percentage (e.g. Price + 20% VAT = 120% = 1.20). 2. Divide final price by multiplier (Price ÷ 1.20).



DO NOT CALCULATE 20% OF £144!

Original = £144 ÷ 1.20 = £120

⋆ Golden Rule: To go BACKWARDS to the original, DIVIDE by the multiplier!

⚠ Watch Out: Never calculate the percentage of the sale/final price and subtract! 20% of £144 is £28.80 (£115.20 = WRONG).

10. Simple vs Compound Interest

NUMBER SYSTEM

Fixed Amount vs Interest on Interest

Simple: Same fixed interest paid every year. Compound: Total = Principal × (Multiplier)^Years. Interest is earned on accumulated interest.

SIMPLE INTEREST

£1,000 at 5% for 3 yrs

£50 + £50 + £50 = £150

Total = £1,150

COMPOUND INTEREST

£1,000 × $(1.05)^3$

£1,000 × 1.157625

Total = £1,157.63

⋆ Golden Rule: Compound formula: Total = Principal × $(1 + r/100)^n$

⚠ Watch Out: Check whether the question asks for the TOTAL ACCUMULATED AMOUNT or just the INTEREST EARNED!

11. Sharing in a 3-Part Ratio

NUMBER SYSTEM

Add Parts, Divide, Multiply

1. Add all parts together to find total parts (e.g. 2 + 3 + 5 = 10 parts). 2. Divide total amount by total parts to find value of 1 part. 3. Multiply 1 part by each share.

Share £480 in ratio 2 : 3 : 5

1. Total: 10 parts

2. £480÷10 = £48

3. ×2, ×3, ×5

Shares: £96 : £144 : £240 (Sum = £480 ✓)

⋆ Golden Rule: Check your work: Add the final shares together — they must equal the starting total!

⚠ Watch Out: Don't divide the total by just one number! Always sum ALL parts first.

Challenge: Reverse Percentages (Original Value)

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N8

A coat costs £136 in a 15% off sale. What was its original price before the sale?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Sale price = 100% - 15% = 85% = 0.85. Original = £136 ÷ 0.85 = £160.

Challenge: Percentage Change Formula

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N7

A gym membership increases from £35 per month to £42 per month. Calculate the percentage increase.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Change = £42 - £35 = £7. % Increase = (7 ÷ 35) × 100 = 1/5 × 100 = 20%.

Challenge: Sharing in a 3-Part Ratio

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N10

Three partners invest in a business in the ratio 3 : 4 : 5. Profits of £72,000 are shared in this ratio. What is the largest share?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Total parts = 3 + 4 + 5 = 12. Value of 1 part = £72,000 ÷ 12 = £6,000. Largest share (5 parts) = 5 × £6,000 = £30,000.

Challenge: Simple vs Compound Interest

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N9

£4,000 is invested in a bond paying 4% compound interest annually. What is the total value after 3 years?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Multiplier = 1.04. Total = £4,000 × $(1.04)^3$ = £4,000 × 1.124864 = £4,499.46.

12. Reverse Ratios (Difference / One Share)

NUMBER SYSTEM

Find 1 Part from the Difference

When told 'Person A gets £X more than Person B': 1. Find difference in ratio parts. 2. Divide given money by difference in parts = 1 part. 3. Multiply by required share.

Ratio 5 : 3 • Liam gets £40 more than Maya

1. Difference in parts: 5 - 3 = 2 parts
2. Value of 1 part: £40 ÷ 2 = £20
3. Total money: (5 + 3) = 8 parts × £20 = £160

⋆ Golden Rule: Match the given amount to its specific parts, NEVER the total parts!

⚠ Watch Out: Don't divide £40 by 8! The £40 represents the DIFFERENCE (2 parts), not the total.

13. Direct & Inverse Proportion

NUMBER SYSTEM

Direct (Both Up) vs Inverse (One Up, One Down)

Direct: As X doubles, Y doubles (Scale factor or Unitary method). Inverse: As workers increase, time decreases! Workers × Time = Constant Total Work.

DIRECT PROPORTION

4 books cost £18

1 book = 18 ÷ 4 = £4.50

7 books = 7 × 4.5 = £31.50

INVERSE PROPORTION

3 painters take 8 hours

Total = 3 × 8 = 24 hrs

6 painters = 24 ÷ 6 = 4 hrs

⋆ Golden Rule: More workers take LESS time! Multiply workers by hours to get total person-hours.

⚠ Watch Out: Don't scale inverse proportion by multiplying both! If 3 workers take 8 hrs, 6 workers do NOT take 16 hrs!

14. Best Buy & Unit Pricing

NUMBER SYSTEM

Cost per Unit (Price ÷ Quantity)

To compare value for money, find the cost for 1 unit (100g, 1ml, or 1 item): Cost per 100g = (Price in pence ÷ Grams) × 100. The lowest cost per unit is the best value.

Pack A: 500g for £2.40

240p ÷ 5 =

48p per 100g

Pack B: 800g for £3.60

360p ÷ 8 =

45p per 100g ⋆

⋆ Golden Rule: Convert pounds to pence first to avoid difficult decimal calculations!

⚠ Watch Out: Don't just pick the cheaper price tag — a smaller pack might cost more per 100g.

15. Rounding to Significant Figures

NUMBER SYSTEM

Count from First Non-Zero Digit

1st sig fig is the first digit that is NOT zero reading left-to-right. Look at next digit: 5 or more rounds UP, 4 or less rounds DOWN. Fill trailing spaces with placeholder zeroes.

Round 0.004782 to 2 sig figs:

Leading zeros do NOT count! First non-zero = 4 (1st), 7 (2nd).

Next digit is 8 (rounds 7 UP to 8) → 0.0048

Round 48,290 to 1 sig fig: 48,290 → 50,000 (keep placeholder zeroes!)

⋆ Golden Rule: Leading zeros in decimals NEVER count as significant figures!

⚠ Watch Out: Don't turn 48,290 into 5! You must keep the placeholder zeros (50,000).

Challenge: Direct & Inverse Proportion

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N12

4 technicians take 6 hours to test a server system. How long would 3 technicians take working at the same rate?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Total work = 4 × 6 = 24 person-hours. For 3 technicians: 24 ÷ 3 = 8 hours.

Challenge: Reverse Ratios (Difference / One Share)

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N11

In a bakery, loaves to rolls are baked in the ratio 4 : 7. There are 45 more rolls than loaves. How many loaves were baked?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Difference in parts = 7 - 4 = 3 parts. 1 part = 45 ÷ 3 = 15 items. Loaves (4 parts) = 4 × 15 = 60 loaves.

Challenge: Rounding to Significant Figures

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N14

Round: (a) 63,740 to 2 significant figures. (b) 0.05084 to 2 significant figures.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

(a) 63,740 → 64,000 (7 rounds 3 up). (b) 0.05084 → 0.051 (8 rounds 0 up; first non-zero is 5).

Challenge: Best Buy & Unit Pricing

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N13

Cereal Brand X: 450g costs £2.25. Brand Y: 600g costs £2.88. Which is better value for money?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Brand X: 225p ÷ 4.5 = 50p per 100g. Brand Y: 288p ÷ 6 = 48p per 100g. Brand Y is better value by 2p per 100g.

16. Estimation & Approximation

NUMBER SYSTEM

Round Every Number to 1 Sig Fig

To estimate calculations and verify checks in Section A: Round EACH number to 1 significant figure BEFORE performing any multiplication or division.

Estimate: (48.7 × 19.3) ÷ 0.49

1. Round to 1 sig fig: 48.7 → 50, 19.3 → 20, 0.49 → 0.5
2. Calculate numerator: 50 × 20 = 1,000
3. Divide: 1,000 ÷ 0.5 = 2,000 (Dividing by 0.5 doubles!)

⋆ Golden Rule: Dividing by 0.5 is the exact same as multiplying by 2!

⚠ Watch Out: Do NOT calculate the exact answer first and round at the end! Examiners award zero marks if unrounded values are used.

17. Powers, Roots & Indices

NUMBER SYSTEM

Square, Cube & Roots

$x^2 = x \times x$. $x^3 = x \times x \times x$. Square root ($\sqrt{x}$) is inverse of squaring. Cube root ($\sqrt[3]{x}$) is the number multiplied by itself 3 times to give x.

SQUARE ROOTS

$8^2 = 64 \rightarrow \sqrt{64} = 8$

$11^2 = 121 \rightarrow \sqrt{121} = 11$

$\sqrt{144} = 12$

CUBE ROOTS

$2^3 = 2 \times 2 \times 2 = 8 \rightarrow \sqrt[3]{8} = 2$

$3^3 = 3 \times 3 \times 3 = 27 \rightarrow \sqrt[3]{27} = 3$

$5^3 = 125 \rightarrow \sqrt[3]{125} = 5$

⋆ Golden Rule: 5^3 is NOT 5 × 3 = 15! 5^3 is 5 × 5 × 5 = 125.

⚠ Watch Out: Square root of a number between 0 and 1 gets BIGGER! √0.25 = 0.5.

18. Formula Substitution

NUMBER SYSTEM

Replace Letters with Numbers

1. Replace each variable with given numbers (use brackets around negatives!). 2. Follow BIDMAS: Multiply adjacent terms before adding. 3. Check units.

Wage Formula: W = 12h + 1.5(oh)

If regular hours h = 35 and overtime hours oh = 6:

W = (12 × 35) + 1.5 × (12 × 6)

W = £420 + £108 = £528 Total Wage

⋆ Golden Rule: Letters side-by-side mean MULTIPLY: 4ab means 4 × a × b!

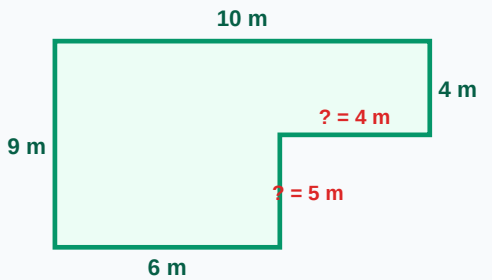
⚠ Watch Out: Squaring a negative number gives a positive: $(-4)^2 = +16$, NOT -16.

19. Perimeter of Composite Shapes

SHAPE & SPACE

Find Missing Sides, Then Add All

1. Find every missing side: Horizontal missing side = Total width - given width. Vertical missing side = Total height - given height. 2. Sum ALL outer sides.



Missing Sides:

Width = 10 - 6 = 4 m

Height = 9 - 4 = 5 m

⋆ Golden Rule: Tick off each side as you add it to avoid counting sides twice or skipping one!

⚠ Watch Out: Never add internal dotted lines when finding perimeter — perimeter is ONLY the outside border!

Challenge: Powers, Roots & Indices

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N16

Evaluate: 4^3 - √169 + 2^4.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

4^3 = 64. √169 = 13. 2^4 = 16. Calculation: 64 - 13 + 16 = 51 + 16 = 67.

Challenge: Estimation & Approximation

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N15

Use approximation to 1 significant figure to estimate: (312 × 4.89) ÷ 0.19.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Round to 1 sf: 300 × 5 ÷ 0.2. 300 × 5 = 1,500. 1,500 ÷ 0.2 = 1,500 × 5 = 7,500.

Challenge: Perimeter of Composite Shapes

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS18

An L-shaped lawn has outer dimensions 12 m by 8 m, with an inset cut of 5 m wide and 3 m deep. What is the perimeter?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Missing horizontal = 12 - 5 = 7 m. Missing vertical = 8 - 3 = 5 m. Perimeter = 12 + 8 + 7 + 3 + 5 + 5 = 40 m.

Challenge: Formula Substitution

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L2.N17

Kinematics: $v^2 = u^2 + 2as$. Find v when u = 3, a = 4, and s = 5.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

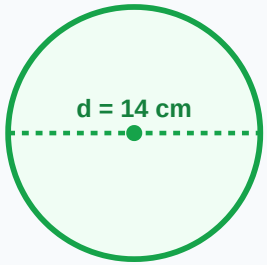
$v^2 = 3^2 + (2 \times 4 \times 5) = 9 + 40 = 49$. v = √49 = 7.

20. Circumference of a Circle

SHAPE & SPACE

Circumference = 3.142 × Diameter

Circumference is the distance around the outside edge of a circle. Diameter (d) goes straight across through center. Radius (r) is from center to edge (d = 2r).



CIRCUMFERENCE

C = 3.142 × Diameter

C = 3.142 × 14

= 43.99 cm

⋆ Golden Rule: Distance around outside: Circumference = 3.142 × Diameter!

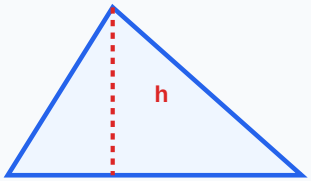
⚠ Watch Out: If given radius (r), you MUST double it before multiplying by 3.142 (Diameter = 2 × r)!

21. Area of Triangle & Parallelogram

SHAPE & SPACE

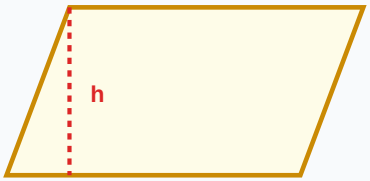
Triangle: 1/2 × b × h | Parallelogram: b × h

Both use PERPENDICULAR vertical height at 90° to the base. Triangle is halved because two identical triangles make a parallelogram/rectangle.



base (b)

Area = (b × h) ÷ 2



base (b)

Area = b × h (NO ÷ 2!)

⋆ Golden Rule: Always use the perpendicular vertical height, NEVER the slanted side!

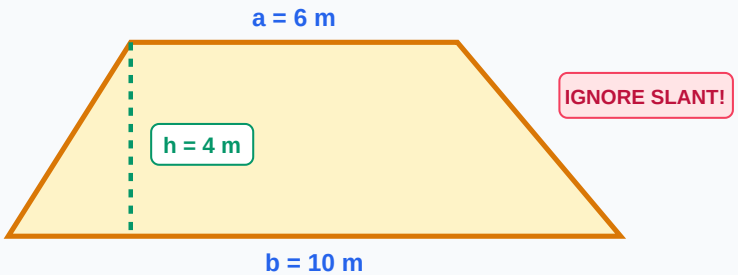
⚠ Watch Out: Don't divide by 2 for a parallelogram — only halve for a triangle!

22. Area of a Trapezium

SHAPE & SPACE

Area = 1/2(a + b)h

1. Add parallel top and bottom sides (a + b). 2. Divide by 2 to get average parallel width. 3. Multiply by perpendicular vertical height (h).



⋆ Golden Rule: Half the sum of parallel sides times vertical height!

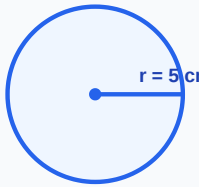
⚠ Watch Out: Examiners include slanted side lengths as a trap. Cross out the slant immediately.

23. Area of a Circle & Semicircle

SHAPE & SPACE

Circle: π r² = 3.142 × r² | Semicircle: (3.142 × r²) ÷ 2

The ² symbol means squared: r² means radius multiplied by itself (r × r). Calculate r² (r × r) first, then multiply by 3.142 (Area = 3.142 × r²). If given diameter, halve it first to get radius. Semicircle area is half the full circle: (3.142 × r²) ÷ 2.



A = 3.142 × 5² = 78.55 cm²



Semicircle Area = (π r²) ÷ 2

(r² means r × r = 5 × 5)

⋆ Golden Rule: Circle Area = 3.142 × r² (where r² means Radius × Radius)!

⚠ Watch Out: Do NOT multiply diameter by itself! Always halve diameter first to find radius.

Challenge: Area of Triangle & Parallelogram

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS20

A triangular roof truss has a base of 8.4 m, perpendicular height of 3.5 m, and sloping sides of 5.5 m. Find its area.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Discard sloping side 5.5 m. Area = 1/2 × base × height = 0.5 × 8.4 × 3.5 = 14.7 m².

Challenge: Circumference of a Circle

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS19

A circular bicycle wheel has a radius of 35 cm. How far does the bicycle travel in 1 full rotation of the wheel? (Use π = 3.142)

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Diameter = 35 × 2 = 70 cm. Circumference = 3.142 × 70 = 219.94 cm (or 2.20 m).

Challenge: Area of a Circle & Semicircle

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS21

A semicircular patio has a straight diameter of 8 metres. Calculate the area of the patio. (Use π = 3.142)

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Radius = 8 ÷ 2 = 4 m. Full circle = 3.142 × 4² = 3.142 × 4 × 4 = 50.272 m². Semicircle = 50.272 ÷ 2 = 25.14 m².

Challenge: Area of a Trapezium

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS20

A pedestrian walkway has parallel edges of 7 m and 11 m, vertical width of 4.5 m, and slant edge of 5.2 m. Calculate its area.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

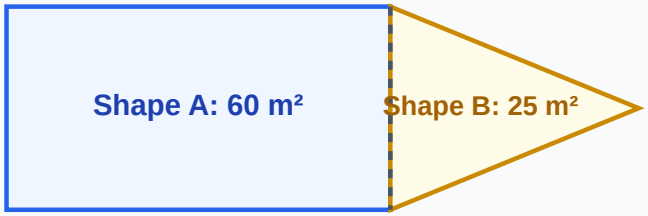
Ignore slant 5.2 m! Area = ((7 + 11) ÷ 2) × 4.5 = 9 × 4.5 = 40.5 m².

24. Area of Composite Shapes

SHAPE & SPACE

Split into Simpler Shapes, Then Add

1. Split compound floor plan into non-overlapping rectangles and triangles with dashed lines. 2. Calculate area of each individual section. 3. Sum all areas.



Total Area = 60 + 25 = 85 m²

⋆ Golden Rule: Draw clear dotted split lines and label areas 'A' and 'B'!

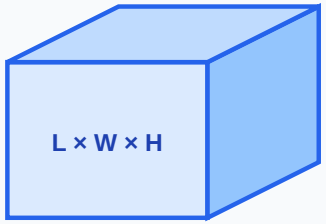
⚠ Watch Out: Check if a section is CUT OUT (e.g. pond in a lawn). If cut out, SUBTRACT its area.

25. Volume of Cuboids & Prisms

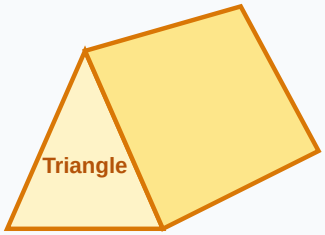
SHAPE & SPACE

Volume = Cross-Sectional Area × Length

A prism has the same 2D shape all the way through: 1. Find area of the front face (cross-section). 2. Multiply by the 3D length/depth. Units are cubic (cm³, m³).



Cuboid



Triangular Prism: Area × Length

⋆ Golden Rule: Always multiply the front face area by the full length/depth!

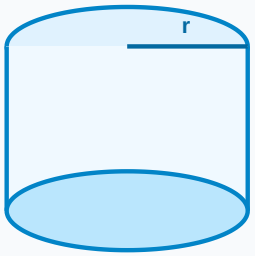
⚠ Watch Out: Don't confuse Volume (3D space in m³) with Surface Area (2D skin in m²).

26. Volume of a Cylinder

SHAPE & SPACE

Volume = π r² h = 3.142 × r² × h

A cylinder is a circular prism: 1. Circular base area = π r² = 3.142 × r² (where r² means r × r). 2. Multiply by vertical height (h): Volume = 3.142 × r² × h. Units are cubic (cm³, m³), where ³ means cubed (x × x × x). 1,000 cm³ = 1 Litre.



CYLINDER FORMULA

V = π r² h = 3.142 × r² × h

(r² means radius × radius)

1,000 cm³ = 1 Litre

⋆ Golden Rule: Cylinder Volume = 3.142 × r² × h (where r² means Radius × Radius)!

⚠ Watch Out: If given diameter, divide by 2 first to get radius! (r = d ÷ 2).

27. Surface Area of a Cuboid

SHAPE & SPACE

SA = 2(lw + lh + wh)

A cuboid has 6 rectangular faces in 3 identical opposite pairs: 1. Front & Back (l × h) × 2. 2. Top & Bottom (l × w) × 2. 3. Left & Right (w × h) × 2. Sum all 6 faces.

Box dimensions: 5 m long, 4 m wide, 3 m high

Top/Bot: 2×(5×4)=40 | Front/Back: 2×(5×3)=30 | Sides: 2×(4×3)=24

Total Surface Area = 40 + 30 + 24 = 94 m²

⋆ Golden Rule: Calculate 3 faces, add them up, then multiply by 2!

⚠ Watch Out: If painting a room, remember to EXCLUDE the floor (or ceiling and windows) if instructed!

Challenge: Volume of Cuboids & Prisms

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS23

A triangular prism has a cross-section triangle with base 6 cm and height 4 cm. Its length is 15 cm. What is its volume?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Triangle area = 1/2 × 6 × 4 = 12 cm². Volume = Cross-section × Length = 12 × 15 = 180 cm³.

Challenge: Area of Composite Shapes

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS22

A warehouse floor is a 15 m by 10 m rectangle with a semicircular loading bay of diameter 6 m extending outwards on one side. What is the total area? (π = 3.14)

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Rectangle = 15 × 10 = 150 m². Semicircle radius = 3 m. Semicircle area = (3.14 × 3²) ÷ 2 = 14.13 m². Total area = 150 + 14.13 = 164.13 m².

Challenge: Surface Area of a Cuboid

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS25

A storage crate with an open top (no lid) is 2 m long, 1.5 m wide, and 1 m high. What is its outside surface area?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Open top = 1 bottom + 2 front/backs + 2 sides. Bottom = 2 × 1.5 = 3 m². Front/Back = 2 × (2 × 1) = 4 m². Ends = 2 × (1.5 × 1) = 3 m². Total SA = 3 + 4 + 3 = 10 m².

Challenge: Volume of a Cylinder

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS24

A hot water tank is a cylinder with diameter 60 cm and height 120 cm. Calculate its capacity in litres (π = 3.14).

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

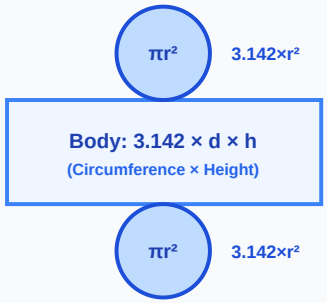
Radius = 60 ÷ 2 = 30 cm. Volume = 3.14 × 30² × 120 = 3.14 × (30 × 30) × 120 = 3.14 × 900 × 120 = 339,120 cm³. Litres = 339,120 ÷ 1,000 = 339.12 L.

28. Surface Area of a Cylinder

SHAPE & SPACE

Total SA = 2πr² + πdh = (2 × 3.142 × r²) + (3.142 × d × h)

A closed cylinder net has 3 parts: Two circular ends (2 × 3.142 × r², where r² means r × r) plus one unwrapped rectangular body (3.142 × d × h).



CLOSED CYLINDER NET

- 2 Ends = 2 × 3.142 × r²
- + Body = 3.142 × d × h

(r² means radius × radius)

⋆ **Golden Rule: Closed Cylinder: 2 circular ends (2 × 3.142 × r²) + curved body (3.142 × d × h)!**

⚠️ **Watch Out: Check if the pipe or can is OPEN at one or both ends — only add circular ends that actually exist.**

29. Metric Unit Conversions

SHAPE & SPACE

Length, Mass & Capacity Ladders

Big unit to small unit = MULTIPLY (×). Small unit to big unit = DIVIDE (÷). Length: km (×1000) m (×100) cm (×10) mm. Mass: tonne (×1000) kg (×1000) g. Volume: L (×1000) mL.

METRIC CONVERSION LADDERS

1 km = 1,000 m • 1 m = 100 cm • 1 cm = 10 mm

1 tonne = 1,000 kg • 1 kg = 1,000 g

1 Litre = 1,000 ml • 1 Litre = 1,000 cm³

⋆ **Golden Rule: Kilo = 1,000 • Centi = 1/100 • Milli = 1/1,000.**

⚠️ **Watch Out: Don't divide 3.5 m by 100 to get cm! When going from metres to cm, you get MORE units (× 100 = 350 cm).**

30. Converting Square & Cubic Units

SHAPE & SPACE

1 m² = 10,000 cm² | 1 m³ = 1,000,000 cm³

To convert square units, apply the linear conversion TWICE (100² = 10,000). To convert cubic units, apply conversion THREE TIMES (100³ = 1,000,000).

SQUARE UNITS (m²)

1 m = 100 cm

1 m² = 100 × 100 cm

1 m² = 10,000 cm²

CUBIC UNITS (m³)

1 m = 100 cm

1 m³ = 100 × 100 × 100

1 m³ = 1,000,000 cm³

⋆ **Golden Rule: Square the conversion factor for area; cube it for volume!**

⚠️ **Watch Out: 1 m² is NOT 100 cm²! That is the most failed unit conversion in Level 2 exams.**

31. Metric to Imperial Approximations

SHAPE & SPACE

Core UK Benchmark Ratios

Memorise standard conversion factors: 5 miles ≈ 8 km (1 mile ≈ 1.6 km). 1 inch ≈ 2.5 cm. 1 kg ≈ 2.2 lbs. 1 gallon ≈ 4.5 litres. 1 foot = 12 inches ≈ 30 cm.

KEY IMPERIAL BENCHMARKS

Distance: 5 miles ≈ 8 km (or 1 mile ≈ 1.6 km)

Weight: 1 kg ≈ 2.2 pounds (lbs)

Liquid: 1 gallon ≈ 4.5 litres • 1 pint ≈ 568 ml

Length: 1 inch ≈ 2.54 cm • 1 foot ≈ 30 cm

⋆ **Golden Rule: 5 miles = 8 km. To convert km to miles, multiply by 5 then divide by 8.**

⚠️ **Watch Out: Kilograms are HEAVIER than pounds! 10 kg is about 22 lbs, NOT 4.5 lbs.**

Challenge: Metric Unit Conversions

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS26

A patient requires 450 mg of medication per dose, 3 times a day. How many grams of medication is taken in 4 days?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Per day = 450 × 3 = 1,350 mg. In 4 days = 1,350 × 4 = 5,400 mg. Grams = 5,400 ÷ 1,000 = 5.4 g.

Challenge: Surface Area of a Cylinder

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS25

A closed tin can has radius 4 cm and height 10 cm. Find total surface area to 1 decimal place (π = 3.142).

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

2 Ends = 2 × 3.142 × 4² = 2 × 3.142 × 16 = 100.54 cm². Curved body = 3.142 × 8 × 10 = 251.36 cm². Total SA = 100.54 + 251.36 = 351.9 cm².

Challenge: Metric to Imperial Approximations

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS28

A UK highway speed limit is 70 mph. A foreign driver's speedometer is in km/h. What is 70 mph in km/h? (5 miles ≈ 8 km)

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

70 ÷ 5 = 14. 14 × 8 = 112 km/h.

Challenge: Converting Square & Cubic Units

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS27

Convert 3.5 m² into cm².

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

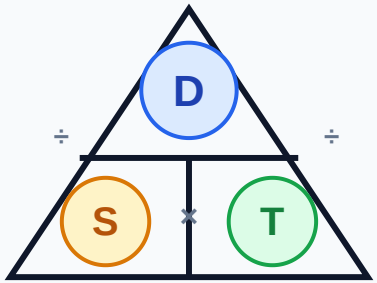
1 m² = 10,000 cm². 3.5 × 10,000 = 35,000 cm².

32. Speed, Distance & Time

SHAPE & SPACE

D = S × T | Decimal Time Warning

Distance = Speed × Time. Speed = Distance ÷ Time. Time = Distance ÷ Speed. Decimal hours must be converted: Multiply decimal minutes by 60 (0.4 hrs = 24 mins).



S-D-T FORMULAS

- Distance = Speed × Time
- Speed = Distance ÷ Time
- Time = Distance ÷ Speed

⚠️ 45m = 0.75h (NEVER 0.45!)

★ **Golden Rule: Minutes to decimal: Divide by 60. Decimal to minutes: Multiply by 60.**

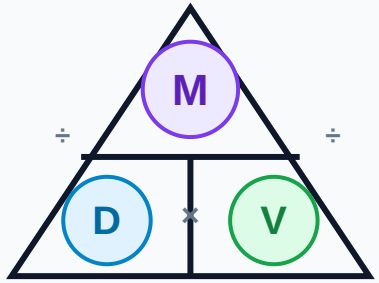
⚠️ **Watch Out: 2 hours 45 minutes on a calculator is 2.75, NEVER 2.45!**

33. Density, Mass & Volume

SHAPE & SPACE

M = D × V (MDV Triangle)

Mass = Density × Volume. Density = Mass ÷ Volume (written in g/cm³ or kg/m³). Volume = Mass ÷ Density. Density measures how tightly matter is packed.



M-D-V FORMULAS

- Mass = Density × Volume
- Density = Mass ÷ Volume
- Volume = Mass ÷ Density

⚖️ g with cm³ | kg with m³

★ **Golden Rule: Remember: 'Department of Motor Vehicles' (D-M-V) or Mass on top!**

⚠️ **Watch Out: Check units: If density is in g/cm³, volume must be in cm³ and mass must be in grams.**

34. Scale Drawings & Maps

SHAPE & SPACE

Scale Ratios (1 : n)

Scale 1 : 50 means 1 cm on paper represents 50 cm in real life (0.5 m). Scale 1 : 25,000 means 1 cm = 250 m. Map to Real: MULTIPLY. Real to Map: DIVIDE.

ARCHITECTURAL: 1:50

1 cm on plan = 50 cm real

50 cm = 0.5 metres

6 cm on plan = 3.0 m

MAP SCALE: 1:25,000

1 cm = 25,000 cm

25,000 cm ÷ 100 = 250 m

4 cm = 1,000 m (1 km)

★ **Golden Rule: Both sides of a scale ratio start in the SAME unit (centimetres)!**

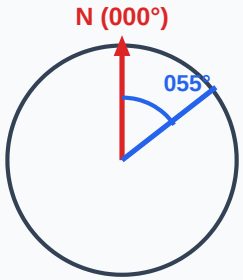
⚠️ **Watch Out: Always convert cm to metres (÷ 100) or km (÷ 100,000) for real-world answers.**

35. Bearings (3-Figure Bearings)

SHAPE & SPACE

North • Clockwise • 3 Figures

1. Measure angle clockwise starting from North line (000°). 2. Always write using THREE digits (e.g. 045°, 090°, 260°). Back bearing = Add or subtract 180°.



★ **Golden Rule: Always 3 digits! 55° must be written as 055°.**

⚠️ **Watch Out: Check which location the bearing is measured FROM: 'Bearing of B from A' means draw North at A!**

Challenge: Density, Mass & Volume

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS30

A solid brass statue has a volume of 450 cm³ and a density of 8.5 g/cm³. What is the mass of the statue in kilograms?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Mass = Density × Volume = 8.5 × 450 = 3,825 grams. In kilograms = 3,825 ÷ 1,000 = 3.825 kg.

Challenge: Speed, Distance & Time

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS29

A train travels 140 miles at an average speed of 56 mph. How long does the journey take in hours and minutes?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Time = Distance ÷ Speed = 140 ÷ 56 = 2.5 hours. 0.5 hours × 60 = 30 minutes. Journey time = 2 hours 30 minutes.

Challenge: Bearings (3-Figure Bearings)

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS32

The bearing of town B from town A is 072°. What is the back-bearing of town A from town B?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Since 072° < 180°, add 180°: Back bearing = 072° + 180° = 252°.

Challenge: Scale Drawings & Maps

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS31

On an architectural blueprint with scale 1 : 200, a sports hall wall measures 8.5 cm. What is the actual length of the wall in metres?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Real length = 8.5 × 200 = 1,700 cm. In metres = 1,700 ÷ 100 = 17 metres.

36. Plans & Elevations

SHAPE & SPACE

Plan View • Front • Side Elevation

Plan View = Looking straight down from the bird's eye view. Front Elevation = Looking straight from front. Side Elevation = Looking straight from side.



⋆ Golden Rule: Draw each 2D projection as if you are standing directly in line with that face!

⚠️ Watch Out: Plan view NEVER shows height — it only shows width and depth.

37. Averages: Mean, Median, Mode & Range

HANDLING DATA

The 4 Core Statistics

Mean = Sum of values ÷ Count. Median = Middle value after ordering low to high. Mode = Most frequent value. Range = Highest - Lowest (spread, NOT an average).

MEAN = Total ÷ Count
Add all up and share equally

MEDIAN = Middle Value
MUST put in order first!

MODE = Most Common
Number that appears most often

RANGE = High - Low
Spread of data (NOT an average!)

⋆ Golden Rule: Range is a measure of consistency/spread, NEVER an average!

⚠️ Watch Out: For the median with an even count of numbers, add the middle two and divide by 2.

38. Mean from a Frequency Table

HANDLING DATA

Mean = Σfx ÷ Σf

1. Multiply value (x) by frequency (f) to make an fx column. 2. Sum the fx column (Total score Σfx). 3. Sum the frequency column (Total count Σf). 4. Divide: Σfx ÷ Σf.

Score (x)	Freq (f)	x × f (fx)
1	3	1 × 3 = 3
2	5	2 × 5 = 10
13 ÷ 8 = 1.625	Total f = 8	Total fx = 13

⋆ Golden Rule: Divide by total frequency (Σf), NEVER by the number of rows in the table!

⚠️ Watch Out: Don't just average the 'x' column — that ignores how many times each value occurred.

39. Estimated Mean (Grouped Data)

HANDLING DATA

Midpoints × Frequency ÷ Total Frequency

1. Find midpoint of each group: (Lower bound + Upper bound) ÷ 2. 2. Multiply midpoint by frequency (mp × f). 3. Sum (mp × f). 4. Divide by total frequency.

Grouped Data: 10 < t ≤ 20 (f = 4)

1. Midpoint = (10 + 20) ÷ 2 = 15

2. mp × f = 15 × 4 = 60

Estimated Mean = Total (mp × f) ÷ Total Frequency

⋆ Golden Rule: It is an 'ESTIMATE' because the exact individual raw values inside the group are unknown!

⚠️ Watch Out: Be careful finding midpoints: for 0 to 10, midpoint is 5. For 20 to 50, midpoint is 35 (not 30).

Challenge: Averages: Mean, Median, Mode & Range

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD34

Find the median and range of this dataset: 14, 8, 22, 19, 14, 27, 11.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Ordered: 8, 11, 14, 14, 19, 22, 27 (7 numbers). Middle is 4th number = 14 (Median). Range = 27 - 8 = 19.

Challenge: Plans & Elevations

SHAPE & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L2.MSS33

A 3D stepped staircase block is 3 cubes wide, 2 cubes deep, and 3 cubes high. Which 2D elevation shows the step pattern?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

The side elevation shows the step profile (staircase outline).

Challenge: Estimated Mean (Grouped Data)

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD36

Times: 0–10 mins (f=2, mp=5), 10–20 mins (f=6, mp=15), 20–30 mins (f=2, mp=25). Calculate estimated mean.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Total (mp × f) = (5×2) + (15×6) + (25×2) = 10 + 90 + 50 = 150. Total f = 2 + 6 + 2 = 10. Estimated Mean = 150 ÷ 10 = 15 mins.

Challenge: Mean from a Frequency Table

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD35

Cars per household: 0 cars (f=4), 1 car (f=10), 2 cars (f=6). Calculate the mean number of cars per household.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

fx: (0×4) + (1×10) + (2×6) = 0 + 10 + 12 = 22. Total frequency Σf = 4 + 10 + 6 = 20. Mean = 22 ÷ 20 = 1.1 cars.

40. Modal Class & Median Group

HANDLING DATA

Identifying Class Intervals

Modal Class = The group interval with the HIGHEST frequency number. Median Group = The interval that contains the middle position ((Total f + 1) ÷ 2).

MODAL CLASS

Look down frequency column

HIGHEST Frequency

MEDIAN GROUP

Find position: (n + 1) ÷ 2

Cumulative Count

⋆ Golden Rule: Always write the ENTIRE interval inequality (e.g. 20 < t ≤ 30), not just one number!

⚠️ Watch Out: The modal class is the interval itself, NOT the frequency count.

41. Comparing Two Data Sets

HANDLING DATA

Average + Spread in Context

To earn full marks on Pearson comparison questions, you MUST state: 1. One Average comparison in context ('Branch A has a higher mean sales by £50'). 2. One Range/Spread comparison ('Branch B is more consistent because its range is smaller').

PEARSON GOLDEN 2-PART COMPARISON

1. AVERAGE: "Group A on average is higher because mean is..."
2. SPREAD: "Group B is more consistent because range is smaller..."

⚠️ Must mention numbers AND the real-world context!

⋆ Golden Rule: Smaller range = more consistent / reliable results!

⚠️ Watch Out: Always write comparison words (higher, lower, more consistent) in context!

42. Probability of Single Events

HANDLING DATA

P(Event) = Favourable ÷ Total

Probability is always between 0 (Impossible) and 1 (Certain). Can be written as a fraction, decimal, or percentage. P(Not event) = 1 - P(Event).

P = Favourable Outcomes ÷ Total



0 (Impossible) 0.5 (Even chance) 1 (Certain)

⋆ Golden Rule: Probabilities can NEVER be greater than 1 or negative!

⚠️ Watch Out: Don't write probability as a ratio (e.g. 1 : 4 is WRONG). Write as a fraction (1/5), decimal (0.2), or % (20%).

43. Combined Independent Events

HANDLING DATA

P(A and B) = P(A) × P(B)

Independent events do not affect each other (e.g. flipping a coin and rolling a die). 'AND' rule = MULTIPLY probabilities. 'OR' rule = ADD probabilities.

P(A and B) = P(A) × P(B)

Flipping Heads AND rolling a 6:

P(Heads) = 1/2 • P(Rolling a 6) = 1/6

Combined: 1/2 × 1/6 = 1/12

⋆ Golden Rule: 'AND' means MULTIPLY; 'OR' means ADD!

⚠️ Watch Out: Don't add when you need both to happen: 1/2 + 1/6 = 4/6 (which is much too high for both happening together).

Challenge: Comparing Two Data Sets

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD38

Shift 1: Mean = 45 units, Range = 12. Shift 2: Mean = 41 units, Range = 5. Compare the two shifts.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1. Shift 1 produced more units on average (mean of 45 vs 41). 2. Shift 2 was more consistent because its range was smaller (range of 5 vs 12).

Challenge: Modal Class & Median Group

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD37

Salaries: £15k–£20k (f=5), £20k–£25k (f=12), £25k–£30k (f=3). State: (a) Modal class, (b) Median group.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

(a) Modal class = £20k–£25k (highest frequency 12). (b) Total f = 20. Middle is 10.5th person. 5 in 1st group, next 12 in 2nd group. Median group = £20k–£25k.

Challenge: Combined Independent Events

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD40

The probability that train A is delayed is 0.2. The probability that train B is delayed is 0.15. What is the probability that BOTH trains are delayed?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

P(Both delayed) = 0.2 × 0.15 = 0.03 (or 3%).

Challenge: Probability of Single Events

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD39

A bag contains 5 red, 7 blue, and 8 green tokens. What is the probability of picking a token that is NOT red? Give as a simplified fraction.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

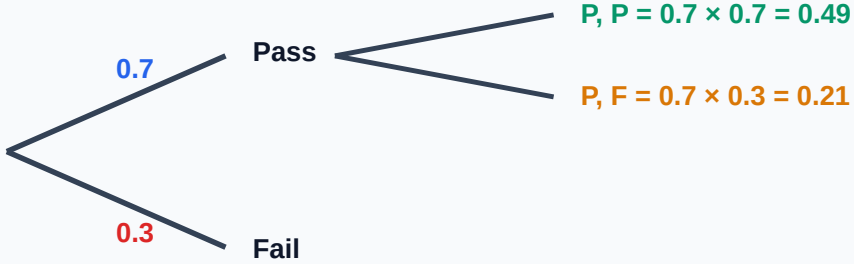
Total tokens = 5 + 7 + 8 = 20. Non-red tokens = 7 + 8 = 15. Probability = 15/20 = 3/4.

44. Probability Tree Diagrams

HANDLING DATA

Multiply Across, Add Down

1. Branches from any single node must add up to 1. 2. MULTIPLY along branches to find outcome probability. 3. ADD individual outcome probabilities together if multiple paths satisfy the question.



⋆ Golden Rule: Every pair of branch splits MUST add up to 1 (100%)!

⚠️ Watch Out: Check whether picking is WITH replacement or WITHOUT replacement (fractions change on 2nd pick!).

45. Two-Way Tables & Probability

HANDLING DATA

Row Totals + Column Totals

1. Every row must add up to its row total. 2. Every column must add up to its column total. 3. Grand total bottom-right must match both row totals and column totals.

Group	Bus	Walk	Total
Men	18	12	30
Women	22	28	50
Total	40	40	80

⋆ Golden Rule: Find the cell with only ONE missing number in its row or column first!

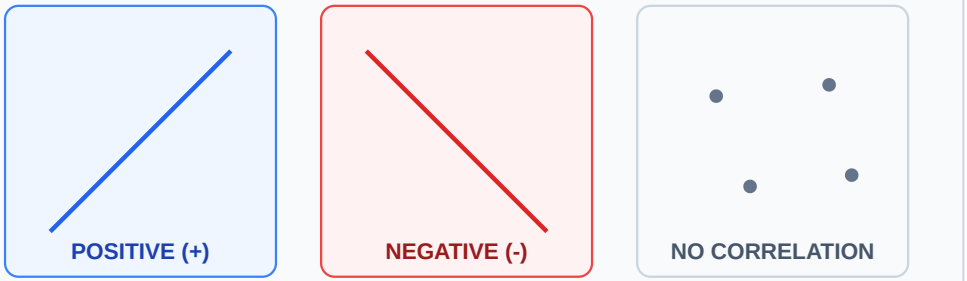
⚠️ Watch Out: For conditional probability ('Given that the person is a woman...'), divide by the WOMEN TOTAL (50), not the grand total (80)!

46. Scatter Diagrams & Correlation

HANDLING DATA

Positive • Negative • Zero Correlation

Positive: As X goes up, Y goes up (slopes uphill). Negative: As X goes up, Y goes down (slopes downhill). Zero: Scattered random cloud with no trend.



⋆ Golden Rule: Draw a line of best fit with an equal balance of points above and below the line!

⚠️ Watch Out: Correlation does NOT prove causation! (e.g. ice cream sales and sunburn both correlate with hot weather).

47. Interpolation vs Extrapolation

HANDLING DATA

Inside Range (Reliable) vs Beyond (Unreliable)

Interpolation: Estimating a value INSIDE the given data range (Reliable). Extrapolation: Extending line of best fit BEYOND the data points (Unreliable / Risky).

INTERPOLATION

Within tested range

e.g. data for 10–50 yrs

RELIABLE PREDICTION

EXTRAPOLATION

Outside data range

Predicting at 90 yrs

UNRELIABLE / RISKY

⋆ Golden Rule: Predictions outside the original data range are UNRELIABLE because trends may flatten or reverse!

⚠️ Watch Out: Examiners will ask 'Why might this prediction be unreliable?' — answer: 'It is extrapolated beyond the data range.'

Challenge: Two-Way Tables & Probability

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD42

In a staff survey of 80 people (30 Men: 18 Bus, 12 Walk; 50 Women: 22 Bus, 28 Walk), what is the probability that a random person is a man who walks to work?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Men who walk = 12. Grand total = 80. Probability = 12/80 = 3/20 (or 0.15 / 15%).

Challenge: Probability Tree Diagrams

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD41

A student has an 80% (0.8) chance of passing Theory, and a 70% (0.7) chance of passing Practical. What is the probability of passing AT LEAST ONE test?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

P(Fails both) = (1 - 0.8) × (1 - 0.7) = 0.2 × 0.3 = 0.06. P(At least one pass) = 1 - 0.06 = 0.94 (or 94%).

Challenge: Interpolation vs Extrapolation

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD44

A company measures sales between 2018 and 2023. They predict sales for 2035 using their line of best fit. Is this reliable?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

No, it is unreliable because 2035 is an extrapolation far beyond the recorded data range, and market conditions may change.

Challenge: Scatter Diagrams & Correlation

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD43

Describe the relationship between temperature and heating fuel consumption: As temperature rises, heating fuel usage drops.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

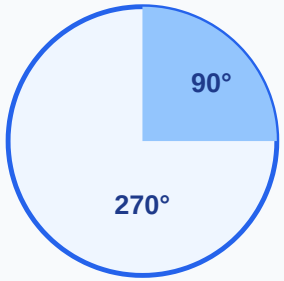
Negative correlation (as temperature increases, heating fuel used decreases).

48. Pie Charts: Angles & Frequencies

HANDLING DATA

Angle = (Value ÷ Total) × 360°

Total degrees in any circle = 360°. Degrees per item = 360° ÷ Total Frequency. To find angle for a category: Multiply its frequency by degrees per item.



Degrees per 1 item:

360° ÷ Total Frequency

e.g. 360° ÷ 72 people = 5° each

⋆ **Golden Rule: Sum of all pie chart sector angles MUST equal exactly 360°!**

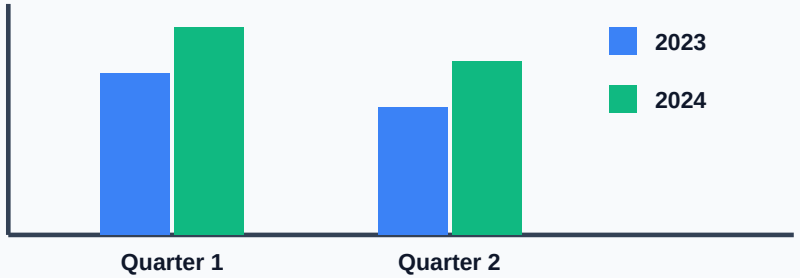
⚠️ **Watch Out: A pie chart only shows PROPORTIONS, not raw quantities, unless the total sample size is given!**

49. Dual & Grouped Bar Charts

HANDLING DATA

Side-by-Side Comparison with Key

Dual bar charts place two bars side-by-side for each category to compare groups (e.g. Male vs Female, 2023 vs 2024). Always check the legend key and vertical axis scale.



⋆ **Golden Rule: Read the key carefully to avoid mixing up years or cohorts!**

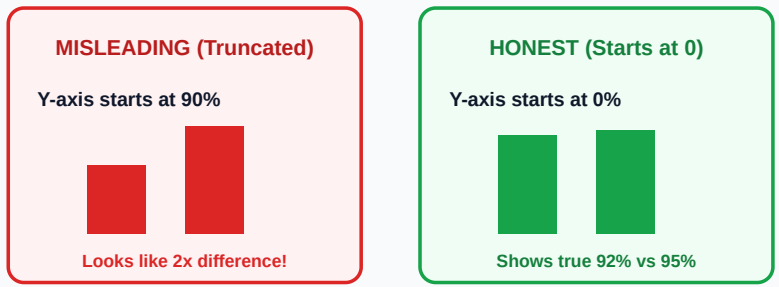
⚠️ **Watch Out: Check the vertical axis scale increments — each grid square may represent 2, 5, 10, or 20 units.**

50. Misleading Graphs & Traps

HANDLING DATA

Non-Zero Baselines & Uneven Scales

Common tricks used to deceive: 1. Broken vertical axis (not starting at 0) exaggerates tiny differences. 2. Unequal scale increments. 3. 3D distortion making front slices look bigger.



⋆ **Golden Rule: Always check if the vertical axis starts at ZERO (0)!**

⚠️ **Watch Out: A graph that looks dramatic often only represents a 1% or 2% actual difference.**

Challenge: Dual & Grouped Bar Charts

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD46

In Quarter 1: 2023 sales were £35k, 2024 sales were £45k. What was the percentage increase from 2023 to 2024?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Increase = £45k - £35k = £10k. % Increase = (10 ÷ 35) × 100 = 28.57% (or 28.6%).

Challenge: Pie Charts: Angles & Frequencies

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD45

In a survey of 120 hospital patients, 45 travelled by bus. What angle should represent bus travel on a pie chart?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Degrees per patient = 360° ÷ 120 = 3° per patient. Bus angle = 45 × 3° = 135°.

Challenge: Misleading Graphs & Traps

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L2.HD47

Explain one way a bar chart comparing company satisfaction could be misleading if one bar appears 4 times taller than another.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

The vertical axis does not start at zero (truncated/broken axis), which exaggerates the visual difference between the bars.